

# scGen Human Validation — Reviewer 2 (Blinded)

## Purpose

Independently determine how each citing paper uses scGen. Machine labels, confidence, agent outputs, and machine-selected evidence are intentionally hidden.

## Workflow

Complete every row in the displayed order. Do not open the other reviewer document or the machine-selection file until both independent reviews are locked.  
Use only the linked approved article versions. Inspect every scGen occurrence and its surrounding methods, results, figures, tables, and available supplements.  
After both reviewers lock their answers, discuss only disagreements in a separate consensus record.

## Allowed values

Review status: DRAFT / LOCKED  
Citation validity and method executed: Yes / No / Unclear  
Uses (select all): Mention only / Benchmark / Extend or develop / Apply to biological data / Other executed use / Unclear  
Primary use: Mention only / Benchmark / Extend or develop / Apply to biological data / Other / Unclear  
Biological insight: Yes / No / Unclear  
Evidence sufficiency: Sufficient / Incomplete / Citation not located / Unclear  
Supporting evidence: shortest exact quote plus section, page, paragraph, figure, or table locator.

## Review table

| Record                               | Article and approved evidence links                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Reviewer 2 independent record                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCGEN-HV-036<br>FORMAL<br>VALIDATION | <b>Learning cell communication from spatial graphs of cells</b><br>DOI: 10.1101/2021.07.11.451750<br>Article page:<br><a href="https://doi.org/10.1101/2021.07.11.451750">https://doi.org/10.1101/2021.07.11.451750</a><br>Full text 1:<br><a href="https://www.biorxiv.org/content/biorxiv/early/2021/07/12/2021.07.11.451750.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2021/07/12/2021.07.11.451750.full.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_653b2dec9c0512787d60.txt:p-0047)                                                                                                                                                                                                                                                                                       | <b>Review status:</b><br><b>Citation validity: Yes</b><br><b>Method executed: No</b><br><b>Uses (select all):</b><br><b>Primary use: Mention only</b><br><b>Biological insight: No</b><br><b>Evidence sufficiency: Yes</b><br><b>Supporting quote + locator:</b><br>Conditional generative models have previously been used to model perturbations of molecular states by small 264 molecules <sup>23</sup> in the context of data from dissociated tissues.<br><br><b>Notes:</b> |
| SCGEN-HV-021<br>FORMAL<br>VALIDATION | <b>Combinatorial prediction of therapeutic perturbations using causally inspired neural networks</b><br>DOI: 10.1038/s41551-025-01481-x<br>Article page:<br><a href="https://doi.org/10.1038/s41551-025-01481-x">https://doi.org/10.1038/s41551-025-01481-x</a><br>Full text 1:<br><a href="https://www.nature.com/articles/s41551-025-01481-x.pdf">https://www.nature.com/articles/s41551-025-01481-x.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_6fd74ea9f2adc84c8130.txt:p-0003);<br>occ-0002<br>(document_6fd74ea9f2adc84c8130.txt:p-0005);<br>occ-0003<br>(document_6fd74ea9f2adc84c8130.txt:p-0006);<br>occ-0004<br>(document_6fd74ea9f2adc84c8130.txt:p-0007);<br>occ-0005<br>(document_6fd74ea9f2adc84c8130.txt:p-0008);<br>occ-0006<br>(document_6fd74ea9f2adc84c8130.txt:p-0008); | <b>Review status:</b><br><b>Citation validity: Yes</b><br><b>Method executed: Yes</b><br><b>Uses (select all):</b><br><b>Primary use: Benchmark</b><br><b>Biological insight: No</b><br><b>Evidence sufficiency: Yes</b><br><b>Supporting quote + locator:</b><br>This feature of PDGrapher enables model training up to 25× faster than indirect prediction methods, such as scGen <sup>22</sup> and CellOT <sup>27</sup> .<br><br><b>Notes:</b>                                 |

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|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>occ-0007<br/> (document_6fd74ea9f2adc84c8130.txt:p-0008);<br/> occ-0008<br/> (document_6fd74ea9f2adc84c8130.txt:p-0009);<br/> occ-0009<br/> (document_6fd74ea9f2adc84c8130.txt:p-0015);<br/> occ-0010<br/> (document_6fd74ea9f2adc84c8130.txt:p-0015);<br/> occ-0011<br/> (document_6fd74ea9f2adc84c8130.txt:p-0015);<br/> occ-0012<br/> (document_6fd74ea9f2adc84c8130.txt:p-0015);<br/> occ-0013<br/> (document_6fd74ea9f2adc84c8130.txt:p-0015);<br/> occ-0014<br/> (document_6fd74ea9f2adc84c8130.txt:p-0015);<br/> occ-0015<br/> (document_6fd74ea9f2adc84c8130.txt:p-0015);<br/> occ-0016<br/> (document_6fd74ea9f2adc84c8130.txt:p-0015);<br/> occ-0017<br/> (document_6fd74ea9f2adc84c8130.txt:p-0016);<br/> occ-0018<br/> (document_6fd74ea9f2adc84c8130.txt:p-0017);<br/> occ-0019<br/> (document_6fd74ea9f2adc84c8130.txt:p-0020);<br/> occ-0020<br/> (document_6fd74ea9f2adc84c8130.txt:p-0020);<br/> occ-0021<br/> (document_6fd74ea9f2adc84c8130.txt:p-0020);<br/> occ-0022<br/> (document_6fd74ea9f2adc84c8130.txt:p-0021);<br/> occ-0023<br/> (document_6fd74ea9f2adc84c8130.txt:p-0021);<br/> occ-0024<br/> (document_6fd74ea9f2adc84c8130.txt:p-0021);<br/> occ-0025<br/> (document_6fd74ea9f2adc84c8130.txt:p-0021);<br/> occ-0026<br/> (document_6fd74ea9f2adc84c8130.txt:p-0021);<br/> occ-0027<br/> (document_6fd74ea9f2adc84c8130.txt:p-0021);<br/> occ-0028<br/> (document_6fd74ea9f2adc84c8130.txt:p-0021);<br/> occ-0029<br/> (document_6fd74ea9f2adc84c8130.txt:p-0021)</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>SCGEN-HV-049</b><br>FORMAL<br>VALIDATION | <p><b>Synthetic single cell RNA sequencing data from small pilot studies using deep generative models</b><br/> DOI: 10.1038/s41598-021-88875-4<br/> Article page:<br/> <a href="https://doi.org/10.1038/s41598-021-88875-4">https://doi.org/10.1038/s41598-021-88875-4</a><br/> Full text 1:<br/> <a href="https://www.nature.com/articles/s41598-021-88875-4.pdf">https://www.nature.com/articles/s41598-021-88875-4.pdf</a><br/> Controller-registered target occurrence locator(s):<br/> occ-0001<br/> (document_d6332622f79e7f2a29c0.txt:p-0010)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p><b>Review status:</b><br/> <b>Citation validity:</b> Yes<br/> <b>Method executed:</b> No<br/> <b>Uses (select all):</b><br/> <b>Primary use:</b> Mention only<br/> <b>Biological insight:</b> No<br/> <b>Evidence sufficiency:</b> Yes<br/> <b>Supporting quote + locator:</b><br/> Furthermore, VAEs were used to generate previously unseen responses to single-cell perturbations based on variables learned in the latent space</p> <p><b>Notes:</b></p> |

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| <b>SCGEN-HV-020</b><br>FORMAL<br>VALIDATION | <b>Domain adaptation for supervised integration of scRNA-seq data</b><br>DOI: 10.1038/s42003-023-04668-7<br>Article page:<br><a href="https://doi.org/10.1038/s42003-023-04668-7">https://doi.org/10.1038/s42003-023-04668-7</a><br>Full text 1:<br><a href="https://www.nature.com/articles/s42003-023-04668-7.pdf">https://www.nature.com/articles/s42003-023-04668-7.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_c05ec999502739b245b9.txt:p-0002)                                                                                                                                                                                                                                                                                        | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> No<br><b>Uses (select all):</b><br><b>Primary use:</b> Mention only<br><b>Biological insight:</b> No<br><b>Evidence sufficiency:</b> Yes<br><b>Supporting quote + locator:</b><br>A few examples of popular methods include canonical correlation analysis (SeuratV2 <sup>3</sup> ), mutual nearest neighbor approach (MNN Correct <sup>4</sup> and fastMNN <sup>5</sup> , Scanorama <sup>6</sup> , BBKNN <sup>7</sup> ), nonnegative matrix factorization (LIGER <sup>8</sup> ), and variational autoencoder (scVI <sup>9</sup> , scGen <sup>10</sup> ).<br><br><b>Notes:</b> |
| <b>SCGEN-HV-007</b><br>FORMAL<br>VALIDATION | <b>Discovering governing equations of biological systems through representation learning and sparse model discovery</b><br>DOI: 10.1093/nargab/lqaf048<br>Article page:<br><a href="https://doi.org/10.1093/nargab/lqaf048">https://doi.org/10.1093/nargab/lqaf048</a><br>Full text 1:<br><a href="https://www.biorxiv.org/content/biorxiv/early/2024/09/23/2024.09.19.613953.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2024/09/23/2024.09.19.613953.full.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_550c6cc33e6cb8e4f260.txt:p-0025)                                                                                                                                                                                        | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> No<br><b>Uses (select all):</b><br><b>Primary use:</b> Mention only<br><b>Biological insight:</b> No<br><b>Evidence sufficiency:</b> Yes<br><b>Supporting quote + locator:</b><br>By carefully designing their architecture and loss function, autoencoders can help extract meaningful representations from complex biological data [ <a href="#">18</a> , <a href="#">26–28</a> ].<br><br><b>Notes:</b>                                                                                                                                                                      |
| <b>SCGEN-HV-032</b><br>FORMAL<br>VALIDATION | <b>Spatial single-cell mass spectrometry defines zonation of the hepatocyte proteome</b><br>DOI: 10.1038/s41592-023-02007-6<br>Article page:<br><a href="https://doi.org/10.1038/s41592-023-02007-6">https://doi.org/10.1038/s41592-023-02007-6</a><br>Full text 1:<br><a href="https://www.biorxiv.org/content/biorxiv/early/2022/12/03/2022.12.03.518957.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2022/12/03/2022.12.03.518957.full.pdf</a><br>Full text 2:<br><a href="https://www.nature.com/articles/s41592-023-02007-6.pdf">https://www.nature.com/articles/s41592-023-02007-6.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_3998cde33dc544a7a33e.txt:p-0030);<br>occ-0002<br>(document_5f32cdfc8029a11592f1.txt:p-0007) | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> No<br><b>Uses (select all):</b><br><b>Primary use:</b> mention only<br><b>Biological insight:</b> no<br><b>Evidence sufficiency:</b> Yes<br><b>Supporting quote + locator:</b><br>Transfer learning might then extend the approach to many new areas, as already shown for single-cell transcriptomics data <sup>24</sup> .<br><br><b>Notes:</b>                                                                                                                                                                                                                               |
| <b>SCGEN-HV-005</b><br>FORMAL<br>VALIDATION | <b>Benchmarking LLM-based agents for single-cell omics analysis</b><br>DOI: 10.1186/s13059-026-03998-z<br>Article page:<br><a href="https://doi.org/10.1186/s13059-026-03998-z">https://doi.org/10.1186/s13059-026-03998-z</a><br>Full text 1: <a href="https://arxiv.org/pdf/2508.13201">https://arxiv.org/pdf/2508.13201</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_3ed8b7d7d2576fa8e251.txt:p-0020);                                                                                                                                                                                                                                                                                                                                        | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> Yes<br><b>Uses (select all):</b><br><b>Primary use:</b> Benchmark<br><b>Biological insight:</b> No<br><b>Evidence sufficiency:</b> Yes                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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|                                                      | <p>occ-0002<br/>(document_3ed8b7d7d2576fa8e251.txt:p-0020);<br/>occ-0003<br/>(document_3ed8b7d7d2576fa8e251.txt:p-0022);<br/>occ-0004<br/>(document_3ed8b7d7d2576fa8e251.txt:p-0063);<br/>occ-0005<br/>(document_3ed8b7d7d2576fa8e251.txt:p-0064);<br/>occ-0006<br/>(document_3ed8b7d7d2576fa8e251.txt:p-0082);<br/>occ-0007<br/>(document_3ed8b7d7d2576fa8e251.txt:p-0100);<br/>occ-0008<br/>(document_3ed8b7d7d2576fa8e251.txt:p-0100);<br/>occ-0009<br/>(document_3ed8b7d7d2576fa8e251.txt:p-0111);<br/>occ-0010<br/>(document_3ed8b7d7d2576fa8e251.txt:p-0111)</p>                                                                                                                                                                                                            | <p><b>Supporting quote + locator:</b>More correlation analysis for cell annotation (CellAssign), perturbation prediction (scGEN), gene imputation (SpaGE) and HVGs selection (Triku) are shown in Additional file 1: Fig. S7b.</p> <p><b>Notes:</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p><b>SCGEN-HV-002</b><br/>FORMAL<br/>VALIDATION</p> | <p><b>Species-agnostic transfer learning for cross-species transcriptomics data integration without gene orthology</b><br/>DOI: 10.1093/bib/bbae004<br/>Article page: <a href="https://doi.org/10.1093/bib/bbae004">https://doi.org/10.1093/bib/bbae004</a><br/>Full text 1: <a href="https://www.biorxiv.org/content/biorxiv/early/2023/08/14/2023.08.11.552752.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2023/08/14/2023.08.11.552752.full.pdf</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_79d3810f985d5821fe1d.txt:p-0003);<br/>occ-0002<br/>(document_79d3810f985d5821fe1d.txt:p-0005);<br/>occ-0003<br/>(document_79d3810f985d5821fe1d.txt:p-0005);<br/>occ-0004<br/>(document_79d3810f985d5821fe1d.txt:p-0034)</p> | <p><b>Review status:</b><br/><b>Citation validity:</b> Yes<br/><b>Method executed:</b> No<br/><b>Uses (select all):</b><br/><b>Primary use:</b> Mention only<br/><b>Biological insight:</b> No<br/><b>Evidence sufficiency:</b> Yes<br/><b>Supporting quote + locator:</b><br/>Previously, scGEN has shown that vector arithmetics on latent space can predict gene signatures of unseen cell types in different species in the lps-stimulated macrophage dataset [35]. Likewise, we analyzed SATL on the same dataset. The stimulation dataset has labels about experimental conditions instead of cell types. Training and testing are done in the same procedure as other single-cell datasets. There are four species and each species has four labels, unst, lps2, lps4 and lps6. Compared to the cross-species analysis of the original work, where only 2336 genes were investigated [17], and the scGEN study, where 6619 genes were investigated [35], our analysis can target all expressed genes from both species, see Materials and methods for details. In this analysis, analyzed genes were 10 000–13 000 genes after single-cell RNA data preprocessing (Table 2).</p> <p><b>Notes:</b></p> |
| <p><b>SCGEN-HV-008</b><br/>FORMAL<br/>VALIDATION</p> | <p><b>Disrupted developmental signaling induces novel transcriptional states</b><br/>DOI: 10.1073/pnas.2418351122<br/>Article page: <a href="https://doi.org/10.1073/pnas.2418351122">https://doi.org/10.1073/pnas.2418351122</a><br/>Full text 1: <a href="https://www.biorxiv.org/content/biorxiv/early/2024/09/06/2024.09.05.610903.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2024/09/06/2024.09.05.610903.full.pdf</a></p>                                                                                                                                                                                                                                                                                                                                      | <p><b>Review status:</b><br/><b>Citation validity:</b> Yes<br/><b>Method executed:</b> No<br/><b>Uses (select all):</b><br/><b>Primary use:</b> Mention only<br/><b>Biological insight:</b> No<br/><b>Evidence sufficiency:</b> Yes<br/><b>Supporting quote + locator:</b>Computational integration techniques can provide a</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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|                                                      | <p>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_2b8424995ee0a401da87.txt:p-0007)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>unified view of all samples by embedding the measured cell states into a common phenotypic space and modeling the perturbation responses of interest, potentially separating them from confounding variation (1–5).</p> <p><b>Notes:</b></p>                                                                                                                                                                                                                                                                                                                     |
| <p><b>SCGEN-HV-028</b><br/>FORMAL<br/>VALIDATION</p> | <p><b>Causal Imputation via Synthetic Interventions</b><br/>DOI: 10.48550/arxiv.2011.03127<br/>Article page:<br/><a href="https://doi.org/10.48550/arxiv.2011.03127">https://doi.org/10.48550/arxiv.2011.03127</a><br/>Full text 1: <a href="https://arxiv.org/pdf/2011.03127">https://arxiv.org/pdf/2011.03127</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_dc94846594a0fc94a1d5.txt:p-0003);<br/>occ-0002<br/>(document_dc94846594a0fc94a1d5.txt:p-0015)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p><b>Review status:</b><br/><b>Citation validity:</b> Yes<br/><b>Method executed:</b> No<br/><b>Uses (select all):</b><br/><b>Primary use:</b> Mention only<br/><b>Biological insight:</b> No<br/><b>Evidence sufficiency:</b> Yes<br/><b>Supporting quote + locator:</b><br/>To overcome this limitation of the fixed effects model, Lotfollahi et al. (2019) introduced the scGen method.</p> <p><b>Notes:</b></p>                                                                                                                                               |
| <p><b>SCGEN-HV-050</b><br/>FORMAL<br/>VALIDATION</p> | <p><b>Deep Learning Based Models for Preimplantation Mouse and Human Development</b><br/>DOI: 10.1101/2024.02.16.580649<br/>Article page:<br/><a href="https://doi.org/10.1101/2024.02.16.580649">https://doi.org/10.1101/2024.02.16.580649</a><br/>Full text 1:<br/><a href="https://www.biorxiv.org/content/biorxiv/early/2024/02/16/2024.02.16.580649.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2024/02/16/2024.02.16.580649.full.pdf</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_9a772d5bb9f72f29304e.txt:p-0005);<br/>occ-0002<br/>(document_9a772d5bb9f72f29304e.txt:p-0005);<br/>occ-0003<br/>(document_9a772d5bb9f72f29304e.txt:p-0006);<br/>occ-0004<br/>(document_9a772d5bb9f72f29304e.txt:p-0007);<br/>occ-0005<br/>(document_9a772d5bb9f72f29304e.txt:p-0015);<br/>occ-0006<br/>(document_9a772d5bb9f72f29304e.txt:p-0015);<br/>occ-0007<br/>(document_9a772d5bb9f72f29304e.txt:p-0015);<br/>occ-0008<br/>(document_9a772d5bb9f72f29304e.txt:p-0019);<br/>occ-0009<br/>(document_9a772d5bb9f72f29304e.txt:p-0024);<br/>occ-0010<br/>(document_9a772d5bb9f72f29304e.txt:p-0025);<br/>occ-0011<br/>(document_9a772d5bb9f72f29304e.txt:p-0026)</p> | <p><b>Review status:</b><br/><b>Citation validity:</b> Yes<br/><b>Method executed:</b> no<br/><b>Uses (select all):</b><br/><b>Primary use:</b> Benchmark<br/><b>Biological insight:</b> No<br/><b>Evidence sufficiency:</b> Yes<br/><b>Supporting quote + locator:</b><br/>Based on a previous study benchmarking integration strategies for scRNA-seq datasets (Luecken et al., 129 2022), we used single-cell variational inference (scVI) (Lopez et al., 2018) and scGen (Lotfollahi et al., 2019) to 130 integrate existing datasets.</p> <p><b>Notes:</b></p> |
| <p><b>SCGEN-HV-017</b><br/>FORMAL<br/>VALIDATION</p> | <p><b>Cells of the adult human heart</b><br/>DOI: 10.1038/s41586-020-2797-4<br/>Article page:<br/><a href="https://doi.org/10.1038/s41586-020-2797-4">https://doi.org/10.1038/s41586-020-2797-4</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p><b>Review status:</b><br/><b>Citation validity:</b> Yes<br/><b>Method executed:</b> Yes<br/><b>Uses (select all):</b><br/><b>Primary use:</b> Apply to Biology data<br/><b>Biological insight:</b> No</p>                                                                                                                                                                                                                                                                                                                                                        |

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|                                      | <p>Full text 1:<br/> <a href="https://www.nature.com/articles/s41586-020-2797-4.pdf">https://www.nature.com/articles/s41586-020-2797-4.pdf</a><br/> Controller-registered target occurrence locator(s):<br/> occ-0001<br/> (document_8fcb457aeb36ed9b0ec0.txt:p-0009);<br/> occ-0002<br/> (document_8fcb457aeb36ed9b0ec0.txt:p-0012);<br/> occ-0003<br/> (document_8fcb457aeb36ed9b0ec0.txt:p-0032)</p>                                                                                                                                                                                                                                                                                | <p><b>Evidence sufficiency:</b> Yes<br/> <b>Supporting quote + locator:</b><br/> Batches were aligned using the batch_correction function from the scGen variational autoencoder<sup>81</sup>.</p> <p><b>Notes:</b></p>                                                                                                                                                                                                                                                                                                                                                                                                   |
| SCGEN-HV-022<br>FORMAL<br>VALIDATION | <p><b>Application and prospects of single-cell and spatial omics technologies in woody plants</b><br/> DOI: 10.48130/fr-2023-0027<br/> Article page:<br/> <a href="https://doi.org/10.48130/fr-2023-0027">https://doi.org/10.48130/fr-2023-0027</a><br/> Full text 1:<br/> <a href="https://www.maxapress.com/data/article/forres/preview/pdf/FR-2023-0027.pdf">https://www.maxapress.com/data/article/forres/preview/pdf/FR-2023-0027.pdf</a><br/> Controller-registered target occurrence locator(s):<br/> occ-0001<br/> (document_4f75272e99730f259da1.txt:p-0008);<br/> occ-0002<br/> (document_4f75272e99730f259da1.txt:p-0037)</p>                                               | <p><b>Review status:</b><br/> <b>Citation validity:</b> Yes<br/> <b>Method executed:</b> No<br/> <b>Uses (select all):</b><br/> <b>Primary use:</b> Mention only<br/> <b>Biological insight:</b> No<br/> <b>Evidence sufficiency:</b> Yes<br/> <b>Supporting quote + locator:</b><br/> For comprehensive analyses and comparisons, it is often necessary to integrate data from different samples or experimental batches, which can be accomplished by software such as Harmony<sup>[53]</sup>, LIGER<sup>[54]</sup>, scMerge<sup>[55]</sup>, scGen<sup>[56]</sup>, and Seurat3<sup>[48]</sup>.</p> <p><b>Notes:</b></p> |
| SCGEN-HV-027<br>FORMAL<br>VALIDATION | <p><b>Single-cell RNA-seq reveals spatially restricted multicellular fibrotic niches during lung fibrosis</b><br/> DOI: 10.1101/569855<br/> Article page: <a href="https://doi.org/10.1101/569855">https://doi.org/10.1101/569855</a><br/> Full text 1:<br/> <a href="https://www.biorxiv.org/content/biorxiv/early/2019/03/06/569855.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2019/03/06/569855.full.pdf</a><br/> No controller-registered target occurrence; independently audit citation location and evidence coverage.</p>                                                                                                                                         | <p><b>Review status:</b><br/> <b>Citation validity:</b> Yes<br/> <b>Method executed:</b> No<br/> <b>Uses (select all):</b><br/> <b>Primary use:</b> Mention only<br/> <b>Biological insight:</b> No<br/> <b>Evidence sufficiency:</b> Yes<br/> <b>Supporting quote + locator:</b><br/> Computational consolidation of single-cell RNA sequencing datasets generated in other models of fibrosis is a promising approach to generate hypotheses with respect to fibrosis pathogenesis that can be tested with causal genetic or pharmacological interventions [39].</p> <p><b>Notes:</b></p>                               |
| SCGEN-HV-047<br>FORMAL<br>VALIDATION | <p><b>Clifti-GPT: privacy-preserving federated fine-tuning and transferable inference of foundation models on clinical single-cell data</b><br/> DOI: 10.1186/s13040-026-00582-w<br/> Article page:<br/> <a href="https://doi.org/10.1186/s13040-026-00582-w">https://doi.org/10.1186/s13040-026-00582-w</a><br/> Full text 1:<br/> <a href="https://link.springer.com/content/pdf/10.1186/s13040-026-00582-w.pdf">https://link.springer.com/content/pdf/10.1186/s13040-026-00582-w.pdf</a><br/> Controller-registered target occurrence locator(s):<br/> occ-0001<br/> (document_230f106da00218449171.txt:p-0017);<br/> occ-0002<br/> (document_230f106da00218449171.txt:p-0017);</p> | <p><b>Review status:</b><br/> <b>Citation validity:</b> Yes<br/> <b>Method executed:</b> Yes<br/> <b>Uses (select all):</b><br/> <b>Primary use:</b> Apply to Biology<br/> <b>Biological insight:</b> No<br/> <b>Evidence sufficiency:</b> Yes<br/> <b>Supporting quote + locator:</b><br/> We applied scGen with default parameters (see Methods) to correct for batch effects, considering all nine batches in the dataset.</p>                                                                                                                                                                                         |

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|                                             | occ-0003<br>(document_230f106da00218449171.txt:p-0025);<br>occ-0004<br>(document_230f106da00218449171.txt:p-0030);<br>occ-0005<br>(document_230f106da00218449171.txt:p-0030);<br>occ-0006<br>(document_230f106da00218449171.txt:p-0036);<br>occ-0007<br>(document_230f106da00218449171.txt:p-0036);<br>occ-0008<br>(document_230f106da00218449171.txt:p-0036)                                                                                                                                                                                                                                                                             | <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>SCGEN-HV-037</b><br>FORMAL<br>VALIDATION | <b>Variational autoencoders learn universal latent representations of metabolomics data</b><br>DOI: 10.1101/2021.01.14.426721<br>Article page:<br><a href="https://doi.org/10.1101/2021.01.14.426721">https://doi.org/10.1101/2021.01.14.426721</a><br>Full text 1:<br><a href="https://www.biorxiv.org/content/biorxiv/early/2021/02/18/2021.01.14.426721.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2021/02/18/2021.01.14.426721.full.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_cefac28e16cb32969c7c.txt:p-0054)                                                             | <b>Review status:</b><br><b>Citation validity: Yes</b><br><b>Method executed: No</b><br><b>Uses (select all):</b><br><b>Primary use:</b><br><b>Biological insight: Mention only</b><br><b>Evidence sufficiency: Yes</b><br><b>Supporting quote + locator:</b><br>The VAE architecture has, for example, proven effective for predicting cell-level response to infection in transcriptomic data not available during training, and predicting drug response from gene expression data where drug response information is sparse <sup>19–22</sup> .<br><br><b>Notes:</b> |
| <b>SCGEN-HV-048</b><br>FORMAL<br>VALIDATION | <b>D-SPIN constructs gene regulatory network models from multiplexed scRNA-seq data revealing organizing principles of cellular perturbation response</b><br>DOI: 10.1101/2023.04.19.537364<br>Article page:<br><a href="https://doi.org/10.1101/2023.04.19.537364">https://doi.org/10.1101/2023.04.19.537364</a><br>Full text 1:<br><a href="https://www.biorxiv.org/content/biorxiv/early/2023/05/20/2023.04.19.537364.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2023/05/20/2023.04.19.537364.full.pdf</a><br>No controller-registered target occurrence;<br>independently audit citation location and evidence coverage. | <b>Review status:</b><br><b>Citation validity: Yes</b><br><b>Method executed: No</b><br><b>Uses (select all):</b><br><b>Primary use: Mention only</b><br><b>Biological insight: No</b><br><b>Evidence sufficiency: Yes</b><br><b>Supporting quote + locator:</b><br>The inclusion of higher-order multi-body interactions or concentration-dependent regulatory responses would enhance accuracy and predictive capabilities [ <a href="#">114</a> , <a href="#">115</a> , <a href="#">116</a> , <a href="#">117</a> ].<br><br><b>Notes:</b>                            |
| <b>SCGEN-HV-019</b><br>FORMAL<br>VALIDATION | <b>Shortcomings of silhouette in single-cell integration benchmarking</b><br>DOI: 10.1038/s41587-025-02743-4<br>Article page:<br><a href="https://doi.org/10.1038/s41587-025-02743-4">https://doi.org/10.1038/s41587-025-02743-4</a><br>Full text 1:<br><a href="https://www.nature.com/articles/s41587-025-02743-4.pdf">https://www.nature.com/articles/s41587-025-02743-4.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_84897ab6cd25226ee536.txt:p-0006)                                                                                                                                      | <b>Review status:</b><br><b>Citation validity: Yes</b><br><b>Method executed: No</b><br><b>Uses (select all):</b><br><b>Primary use: Mention only</b><br><b>Biological insight: No</b><br><b>Evidence sufficiency: yes</b><br><b>Supporting quote + locator:</b><br>From 2017 onward, silhouette-based metrics have also been used for scoring batch effect removal <sup><a href="#">5</a>,<a href="#">7</a>,<a href="#">8</a>,<a href="#">9</a></sup> .<br><br><b>Notes:</b>                                                                                           |
| <b>SCGEN-HV-035</b>                         | <b>lute: estimating the cell composition of heterogeneous tissue with varying cell sizes using gene expression</b><br>DOI: 10.1186/s12864-025-11508-x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Review status:</b><br><b>Citation validity: yes</b><br><b>Method executed: No</b><br><b>Uses (select all):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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| FORMAL<br>VALIDAT<br>ION                     | <p>Article page:<br/><a href="https://doi.org/10.1186/s12864-025-11508-x">https://doi.org/10.1186/s12864-025-11508-x</a><br/>Full text 1:<br/><a href="https://www.biorxiv.org/content/biorxiv/early/2024/04/06/2024.04.04.588105.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2024/04/06/2024.04.04.588105.full.pdf</a><br/>Full text 2:<br/><a href="https://bmcmgenomics.biomedcentral.com/counter/pdf/10.1186/s12864-025-11508-x">https://bmcmgenomics.biomedcentral.com/counter/pdf/10.1186/s12864-025-11508-x</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_b94abc966b9c57ca4cb9.txt:p-0025);<br/>occ-0002<br/>(document_ee70a2ea12d279c619be.txt:p-0019)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p><b>Primary use: Mention only</b><br/><b>Biological insight: No</b><br/><b>Evidence sufficiency: yes</b><br/><b>Supporting quote + locator:</b><br/>However, each of these tools were built on different frameworks with non-uniform input data formats while addressing different types of systematic errors or unwanted bias [<a href="#">19</a>,<a href="#">20</a>,<a href="#">21</a>,<a href="#">22</a>,<a href="#">23</a>,<a href="#">24</a>,<a href="#">25</a>,<a href="#">26</a>].</p> <p><b>Notes:</b></p> |
| SCGEN-<br>HV-018<br>FORMAL<br>VALIDAT<br>ION | <p><b>Causal identification of single-cell experimental perturbation effects with CINEMA-OT</b><br/>DOI: 10.1038/s41592-023-02040-5<br/>Article page:<br/><a href="https://doi.org/10.1038/s41592-023-02040-5">https://doi.org/10.1038/s41592-023-02040-5</a><br/>Full text 1:<br/><a href="https://www.biorxiv.org/content/biorxiv/early/2022/08/01/2022.07.31.502173.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2022/08/01/2022.07.31.502173.full.pdf</a><br/>Full text 2:<br/><a href="https://www.nature.com/articles/s41592-023-02040-5.pdf">https://www.nature.com/articles/s41592-023-02040-5.pdf</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_5dd0fc96859316055ef2.txt:p-0023);<br/>occ-0002<br/>(document_469515c40259229a09e3.txt:p-0005);<br/>occ-0003<br/>(document_469515c40259229a09e3.txt:p-0006);<br/>occ-0004<br/>(document_469515c40259229a09e3.txt:p-0006);<br/>occ-0005<br/>(document_469515c40259229a09e3.txt:p-0006);<br/>occ-0006<br/>(document_469515c40259229a09e3.txt:p-0006);<br/>occ-0007<br/>(document_469515c40259229a09e3.txt:p-0006);<br/>occ-0008<br/>(document_469515c40259229a09e3.txt:p-0006);<br/>occ-0009<br/>(document_469515c40259229a09e3.txt:p-0007);<br/>occ-0010<br/>(document_469515c40259229a09e3.txt:p-0007);<br/>occ-0011<br/>(document_469515c40259229a09e3.txt:p-0007);<br/>occ-0012<br/>(document_469515c40259229a09e3.txt:p-0011);<br/>occ-0013<br/>(document_469515c40259229a09e3.txt:p-0013);<br/>occ-0014<br/>(document_469515c40259229a09e3.txt:p-0013);<br/>occ-0015<br/>(document_469515c40259229a09e3.txt:p-0013);<br/>occ-0016<br/>(document_469515c40259229a09e3.txt:p-0013);<br/>occ-0017<br/>(document_469515c40259229a09e3.txt:p-0013);<br/>occ-0018<br/>(document_469515c40259229a09e3.txt:p-0013);</p> | <p><b>Review status:</b><br/><b>Citation validity: Yes</b><br/><b>Method executed: yes</b><br/><b>Uses (select all):</b><br/><b>Primary use: Benchmark</b><br/><b>Biological insight: No</b><br/><b>Evidence sufficiency: Yes</b><br/><b>Supporting quote + locator:</b><br/>We compared CINEMA-OT with Mixscape, scGen, CPA, ContrastiveVI, and CellOT in our experiments, covering both the default model (cell-type-unaware) and cell-type-aware models for scGen and CPA.</p> <p><b>Notes:</b></p>               |

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|                                                 | occ-0019<br>(document_469515c40259229a09e3.txt:p-0014);<br>occ-0020<br>(document_469515c40259229a09e3.txt:p-0014);<br>occ-0021<br>(document_469515c40259229a09e3.txt:p-0014);<br>occ-0022<br>(document_469515c40259229a09e3.txt:p-0017);<br>occ-0023<br>(document_469515c40259229a09e3.txt:p-0018)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>SCGEN-HV-026</b><br>FORMAL<br>VALIDAT<br>ION | <b>The Power of Two: integrating deep diffusion models and variational autoencoders for single-cell transcriptomics analysis</b><br>DOI: 10.1101/2023.04.13.536789<br>Article page:<br><a href="https://doi.org/10.1101/2023.04.13.536789">https://doi.org/10.1101/2023.04.13.536789</a><br>Full text 1:<br><a href="https://www.biorxiv.org/content/biorxiv/early/2023/04/16/2023.04.13.536789.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2023/04/16/2023.04.13.536789.full.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_e6092864959832ed9def.txt:p-0024);<br>occ-0002<br>(document_e6092864959832ed9def.txt:p-0024);<br>occ-0003<br>(document_e6092864959832ed9def.txt:p-0024);<br>occ-0004<br>(document_e6092864959832ed9def.txt:p-0024);<br>occ-0005<br>(document_e6092864959832ed9def.txt:p-0026);<br>occ-0006<br>(document_e6092864959832ed9def.txt:p-0026);<br>occ-0007<br>(document_e6092864959832ed9def.txt:p-0026);<br>occ-0008<br>(document_e6092864959832ed9def.txt:p-0058);<br>occ-0009<br>(document_e6092864959832ed9def.txt:p-0058);<br>occ-0010<br>(document_e6092864959832ed9def.txt:p-0058);<br>occ-0011<br>(document_e6092864959832ed9def.txt:p-0058);<br>occ-0012<br>(document_e6092864959832ed9def.txt:p-0060);<br>occ-0013<br>(document_e6092864959832ed9def.txt:p-0062) | <b>Review status:</b><br><b>Citation validity: Yes</b><br><b>Method executed: Yes</b><br><b>Uses (select all):</b><br><b>Primary use: Benchmark</b><br><b>Biological insight: No</b><br><b>Evidence sufficiency: Yes</b><br><b>Supporting quote + locator:</b><br>We also conduct a comparison between scVAEDer and the state-of-the-art method scGen (which is based on VAE) by evaluating the correlation values between the predicted average gene expression and real data for T-cell across all genes ( <b>Fig. 4c,d</b> respectively).<br><br><b>Notes:</b> |
| <b>SCGEN-HV-040</b><br>FORMAL<br>VALIDAT<br>ION | <b>Differences between in vivo and ex vivo models drive divergent perturbation outcomes</b><br>DOI: 10.1101/2025.09.02.673692<br>Article page:<br><a href="https://doi.org/10.1101/2025.09.02.673692">https://doi.org/10.1101/2025.09.02.673692</a><br>Full text 1:<br><a href="https://www.biorxiv.org/content/biorxiv/early/2025/09/07/2025.09.02.673692.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2025/09/07/2025.09.02.673692.full.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_9dd3e890037d378b8400.txt:p-0055)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Review status:</b><br><b>Citation validity: No</b><br><b>Method executed:</b><br><b>Uses (select all):</b><br><b>Primary use:</b><br><b>Biological insight:</b><br><b>Evidence sufficiency:</b><br><b>Supporting quote + locator:</b><br><br><b>Notes:</b>                                                                                                                                                                                                                                                                                                     |
| <b>SCGEN-HV-034</b><br>FORMAL<br>VALIDAT<br>ION | <b>Detecting subtle transcriptomic perturbations induced by lncRNAs knock-down in single-cell CRISPRi screening using a new sparse supervised autoencoder neural network</b><br>DOI: 10.3389/fbinf.2024.1340339                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Review status:</b><br><b>Citation validity: Yes</b><br><b>Method executed: No</b><br><b>Uses (select all):</b><br><b>Primary use: Mention only</b>                                                                                                                                                                                                                                                                                                                                                                                                             |

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|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | <p>Article page:<br/> <a href="https://doi.org/10.3389/fbinf.2024.1340339">https://doi.org/10.3389/fbinf.2024.1340339</a><br/> Full text 1:<br/> <a href="https://www.biorxiv.org/content/biorxiv/early/2023/07/11/2023.07.11.548494.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2023/07/11/2023.07.11.548494.full.pdf</a><br/> Full text 2:<br/> <a href="https://hal.science/hal-04795816v1/file/Truchi_Frontiers_Bioinfo24.pdf">https://hal.science/hal-04795816v1/file/Truchi_Frontiers_Bioinfo24.pdf</a><br/> Controller-registered target occurrence locator(s):<br/> occ-0001<br/> (document_d9f816edbc1045f84712.txt:p-0018)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p><b>Biological insight:</b> No<br/> <b>Evidence sufficiency:</b> Yes<br/> <b>Supporting quote + locator:</b><br/> for example, VAEs have been applied to scRNA-seq to predict cell response to biological perturbations (Lotfollahi et al., 2019) and Omics dataset (Eltager et al., 2023).</p> <p><b>Notes:</b></p>                                                                                                                                       |
| <p><b>SCGEN-HV-004</b><br/> <b>FORMAL VALIDATION</b></p> | <p><b>scDREAMER for atlas-level integration of single-cell datasets using deep generative model paired with adversarial classifier</b><br/> DOI: 10.1038/s41467-023-43590-8<br/> Article page:<br/> <a href="https://doi.org/10.1038/s41467-023-43590-8">https://doi.org/10.1038/s41467-023-43590-8</a><br/> Full text 1:<br/> <a href="https://www.nature.com/articles/s41467-023-43590-8.pdf">https://www.nature.com/articles/s41467-023-43590-8.pdf</a><br/> Controller-registered target occurrence locator(s):<br/> occ-0001<br/> (document_ee8eb663889e8c2359cf.txt:p-0008);<br/> occ-0002<br/> (document_ee8eb663889e8c2359cf.txt:p-0008);<br/> occ-0003<br/> (document_ee8eb663889e8c2359cf.txt:p-0008);<br/> occ-0004<br/> (document_ee8eb663889e8c2359cf.txt:p-0008);<br/> occ-0005<br/> (document_ee8eb663889e8c2359cf.txt:p-0008);<br/> occ-0006<br/> (document_ee8eb663889e8c2359cf.txt:p-0010);<br/> occ-0007<br/> (document_ee8eb663889e8c2359cf.txt:p-0010);<br/> occ-0008<br/> (document_ee8eb663889e8c2359cf.txt:p-0010);<br/> occ-0009<br/> (document_ee8eb663889e8c2359cf.txt:p-0010);<br/> occ-0010<br/> (document_ee8eb663889e8c2359cf.txt:p-0010);<br/> occ-0011<br/> (document_ee8eb663889e8c2359cf.txt:p-0010);<br/> occ-0012<br/> (document_ee8eb663889e8c2359cf.txt:p-0010);<br/> occ-0013<br/> (document_ee8eb663889e8c2359cf.txt:p-0010);<br/> occ-0014<br/> (document_ee8eb663889e8c2359cf.txt:p-0011);<br/> occ-0015<br/> (document_ee8eb663889e8c2359cf.txt:p-0011);<br/> occ-0016<br/> (document_ee8eb663889e8c2359cf.txt:p-0012);<br/> occ-0017<br/> (document_ee8eb663889e8c2359cf.txt:p-0012);<br/> occ-0018<br/> (document_ee8eb663889e8c2359cf.txt:p-0012);<br/> occ-0019<br/> (document_ee8eb663889e8c2359cf.txt:p-0012);<br/> occ-0020<br/> (document_ee8eb663889e8c2359cf.txt:p-0012);<br/> occ-0021<br/> (document_ee8eb663889e8c2359cf.txt:p-0014);</p> | <p><b>Review status:</b><br/> <b>Citation validity:</b> Yes<br/> <b>Method executed:</b> No<br/> <b>Uses (select all):</b><br/> <b>Primary use:</b> Benchmark<br/> <b>Biological insight:</b> No<br/> <b>Evidence sufficiency:</b> yes<br/> <b>Supporting quote + locator:</b><br/> f iLISI and cLISI values between scGEN, scANVI, and scDREAMER-Sup for different percentages of missing cell type labels for lung atlas dataset.</p> <p><b>Notes:</b></p> |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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|                                      | occ-0022<br>(document_ee8eb663889e8c2359cf.txt:p-0014);<br>occ-0023<br>(document_ee8eb663889e8c2359cf.txt:p-0018)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| SCGEN-HV-042<br>FORMAL<br>VALIDATION | <b>Embeddings from language models are good learners for single-cell data analysis</b><br>DOI: 10.1016/j.patter.2025.101431<br>Article page:<br><a href="https://doi.org/10.1016/j.patter.2025.101431">https://doi.org/10.1016/j.patter.2025.101431</a><br>Full text 1:<br><a href="https://www.researchsquare.com/article/rs-4014784/latest.pdf">https://www.researchsquare.com/article/rs-4014784/latest.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_9023646e3bd30df62acb.txt:p-0025)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> No<br><b>Uses (select all):</b><br><b>Primary use:</b> Mention only<br><b>Biological insight:</b> No<br><b>Evidence sufficiency:</b> Yes<br><b>Supporting quote + locator:</b><br>To this end, we focused on three distinct tasks spanning both chemical perturbations <sup>104</sup> and gene-level perturbations (e.g., CRISPR-based perturb-seq). <sup>105</sup><br><br><b>Notes:</b>                                                                                                                                                                                                                                                             |
| SCGEN-HV-012<br>FORMAL<br>VALIDATION | <b>scVAE: variational auto-encoders for single-cell gene expression data.</b><br>DOI: 10.1093/bioinformatics/btaa293<br>Article page:<br><a href="https://doi.org/10.1093/bioinformatics/btaa293">https://doi.org/10.1093/bioinformatics/btaa293</a><br>Full text 1:<br><a href="https://www.biorxiv.org/content/biorxiv/early/2019/10/02/318295.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2019/10/02/318295.full.pdf</a><br>No controller-registered target occurrence; independently audit citation location and evidence coverage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> No<br><b>Uses (select all):</b><br><b>Primary use:</b> Mention only<br><b>Biological insight:</b> No<br><b>Evidence sufficiency:</b> Yes<br><b>Supporting quote + locator:</b><br>As mentioned above, it should also be possible to combine encodings of the cells in the latent space and produce in-between cells like <a href="#">Lotfollahi et al. (2018)</a> .<br><br><b>Notes:</b>                                                                                                                                                                                                                                                             |
| SCGEN-HV-039<br>FORMAL<br>VALIDATION | <b>A deep generative model for deciphering cellular dynamics and in silico drug discovery in complex diseases</b><br>DOI: 10.1038/s41551-025-01423-7<br>Article page:<br><a href="https://doi.org/10.1038/s41551-025-01423-7">https://doi.org/10.1038/s41551-025-01423-7</a><br>Full text 1:<br><a href="https://digitalcommons.library.tmc.edu/cgi/viewcontent.cgi?article=7661&amp;context=baylor_docs">https://digitalcommons.library.tmc.edu/cgi/viewcontent.cgi?article=7661&amp;context=baylor_docs</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_b0f410bbf326127d9bce.txt:p-0006);<br>occ-0002<br>(document_b0f410bbf326127d9bce.txt:p-0014);<br>occ-0003<br>(document_b0f410bbf326127d9bce.txt:p-0016);<br>occ-0004<br>(document_b0f410bbf326127d9bce.txt:p-0016);<br>occ-0005<br>(document_b0f410bbf326127d9bce.txt:p-0016);<br>occ-0006<br>(document_b0f410bbf326127d9bce.txt:p-0016);<br>occ-0007<br>(document_b0f410bbf326127d9bce.txt:p-0016);<br>occ-0008<br>(document_b0f410bbf326127d9bce.txt:p-0016);<br>occ-0009 | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> Yes<br><b>Uses (select all):</b><br><b>Primary use:</b> Benchmark<br><b>Biological insight:</b> No<br><b>Evidence sufficiency:</b> Yes<br><b>Supporting quote + locator:</b><br>To demonstrate UNAGI's advantages over existing methods in understanding the dynamics of diseases, we benchmarked it against established methods, including scVI <sup>19</sup> , GraphSCC <sup>22</sup> , scGEN <sup>33</sup> , scGGAN <sup>20</sup> , scGPT <sup>31</sup> , Geneformer <sup>144</sup> , scGNN <sup>21</sup> , Seurat <sup>16</sup> and SCANPY <sup>17</sup> , on the IPF dataset and scRNA COVID-19 PBMC data <sup>128</sup> .<br><br><b>Notes:</b> |

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|                                                      | <p>(document_b0f410bbf326127d9bce.txt:p-0017);<br/>occ-0010<br/>(document_b0f410bbf326127d9bce.txt:p-0017);<br/>occ-0011<br/>(document_b0f410bbf326127d9bce.txt:p-0017);<br/>occ-0012<br/>(document_b0f410bbf326127d9bce.txt:p-0017);<br/>occ-0013<br/>(document_b0f410bbf326127d9bce.txt:p-0017);<br/>occ-0014<br/>(document_b0f410bbf326127d9bce.txt:p-0017);<br/>occ-0015<br/>(document_b0f410bbf326127d9bce.txt:p-0023);<br/>occ-0016<br/>(document_b0f410bbf326127d9bce.txt:p-0023);<br/>occ-0017<br/>(document_b0f410bbf326127d9bce.txt:p-0023);<br/>occ-0018<br/>(document_b0f410bbf326127d9bce.txt:p-0023);<br/>occ-0019<br/>(document_b0f410bbf326127d9bce.txt:p-0023);<br/>occ-0020<br/>(document_b0f410bbf326127d9bce.txt:p-0024);<br/>occ-0021<br/>(document_b0f410bbf326127d9bce.txt:p-0024);<br/>occ-0022<br/>(document_b0f410bbf326127d9bce.txt:p-0024);<br/>occ-0023<br/>(document_b0f410bbf326127d9bce.txt:p-0025);<br/>occ-0024<br/>(document_b0f410bbf326127d9bce.txt:p-0034)</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p><b>SCGEN-HV-044</b><br/>FORMAL<br/>VALIDATION</p> | <p><b>Contrastive Mixture of Posteriors for Counterfactual Inference, Data Integration and Fairness</b><br/>DOI: 10.48550/arxiv.2106.08161<br/>Article page:<br/><a href="https://doi.org/10.48550/arxiv.2106.08161">https://doi.org/10.48550/arxiv.2106.08161</a><br/>Full text 1: <a href="https://arxiv.org/pdf/2106.08161">https://arxiv.org/pdf/2106.08161</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_9ba19ba18f3ae92f6ff2.txt:p-0008);<br/>occ-0002<br/>(document_9ba19ba18f3ae92f6ff2.txt:p-0008);<br/>occ-0003<br/>(document_9ba19ba18f3ae92f6ff2.txt:p-0011)</p>                                                                                                                                                                                                                                                                                                                                                                                | <p><b>Review status:</b><br/><b>Citation validity: Yes</b><br/><b>Method executed: No</b><br/><b>Uses (select all):</b><br/><b>Primary use: Mention only</b><br/><b>Biological insight: no</b><br/><b>Evidence sufficiency: Yes</b><br/><b>Supporting quote + locator:</b><br/>Predicting cell-level response to intervention has been previously tackled using VAEs by scGen (Lotfollahi et al., 2019b) and trVAE (Lotfollahi et al., 2019a). Although not always referred to as such, this problem is an example of counterfactual inference. Both scGen and trVAE can be interpreted as approaching counterfactual inference by enforcing alignment between different conditions in latent space with a linear translation and MMD penalty respectively.</p> <p><b>Notes:</b></p> |
| <p><b>SCGEN-HV-043</b><br/>FORMAL<br/>VALIDATION</p> | <p><b>Technical and legal aspects of federated learning in bioinformatics: applications, challenges and opportunities</b><br/>DOI: 10.3389/fdgth.2025.1644291<br/>Article page:<br/><a href="https://doi.org/10.3389/fdgth.2025.1644291">https://doi.org/10.3389/fdgth.2025.1644291</a><br/>Full text 1:<br/><a href="https://public-pages-files-2025.frontiersin.org/journals/digital-health/articles/10.3389/fdgth.2025.1644291/pdf">https://public-pages-files-2025.frontiersin.org/journals/digital-health/articles/10.3389/fdgth.2025.1644291/pdf</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_0164d35eae5262e19dc3.txt:p-0009);</p>                                                                                                                                                                                                                                                                                                                  | <p><b>Review status:</b><br/><b>Citation validity: Yes</b><br/><b>Method executed: No</b><br/><b>Uses (select all):</b><br/><b>Primary use: Extend or Develop</b><br/><b>Biological insight: No</b><br/><b>Evidence sufficiency: Yes</b><br/><b>Supporting quote + locator:</b><br/>More recently, Bakhtiari et al. (93) introduced FedscGen, a federated implementation of scGen (94), a variational autoencoder-based method for batch effect correction. FedscGen</p>                                                                                                                                                                                                                                                                                                             |

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|                                             | occ-0002<br>(document_0164d35eae5262e19dc3.txt:p-0009);<br>occ-0003<br>(document_0164d35eae5262e19dc3.txt:p-0009);<br>occ-0004<br>(document_0164d35eae5262e19dc3.txt:p-0016);<br>occ-0005<br>(document_0164d35eae5262e19dc3.txt:p-0016)                                                                                                                                                                                                                                                                                                                                                      | employs secure SMPC for privacy-preserving aggregation and achieves results that closely match those obtained under centralised training.<br><br><b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>SCGEN-HV-024</b><br>FORMAL<br>VALIDATION | <b>Assessing and mitigating batch effects in large-scale omics studies</b><br>DOI: 10.1186/s13059-024-03401-9<br>Article page:<br><a href="https://doi.org/10.1186/s13059-024-03401-9">https://doi.org/10.1186/s13059-024-03401-9</a><br>Full text 1:<br><a href="https://link.springer.com/content/pdf/10.1186/s13059-024-03401-9.pdf">https://link.springer.com/content/pdf/10.1186/s13059-024-03401-9.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_d011dcd0741d704029f6.txt:p-0028);<br>occ-0002<br>(document_d011dcd0741d704029f6.txt:p-0040) | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> No<br><b>Uses (select all):</b><br><b>Primary use:</b> Benchmark<br><b>Biological insight:</b> No<br><b>Evidence sufficiency:</b> Yes<br><b>Supporting quote + locator:</b><br>DL methods usually use neural network algorithms to identify and remove batch effects from the data, including AutoClass [141], DESC [142], scGen [143], scVI [144], and so on.<br><br><b>Notes:</b>                                                                                                                                                                                                                                                |
| <b>SCGEN-HV-045</b><br>FORMAL<br>VALIDATION | <b>scCompressSA: dual-channel self-attention based deep autoencoder model for single-cell clustering by compressing gene–gene interactions</b><br>DOI: 10.1186/s12864-024-10286-2<br>Article page:<br><a href="https://doi.org/10.1186/s12864-024-10286-2">https://doi.org/10.1186/s12864-024-10286-2</a><br>Full text 1:<br><a href="https://www.researchsquare.com/article/rs-3674838/latest.pdf">https://www.researchsquare.com/article/rs-3674838/latest.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_62463667ea5a15b4ae77.txt:p-0018)        | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> No<br><b>Uses (select all):</b><br><b>Primary use:</b> Mention only<br><b>Biological insight:</b> No<br><b>Evidence sufficiency:</b> Yes<br><b>Supporting quote + locator:</b><br>As a high-throughput technology, single-cell RNA sequencing (scRNA-seq) make it feasible to investigate the cellular heterogeneity and thus played a crucial role in systems biology and precision medicine. Distributions of cell subpopulations are closely related with cell states and disease subtypes. Cell clustering of scRNA-seq data is crucial to detect meaningful patterns from raw gene count matrix [1,2,3].<br><br><b>Notes:</b> |
| <b>SCGEN-HV-033</b><br>FORMAL<br>VALIDATION | <b>Learning interpretable cellular and gene signature embeddings from single-cell transcriptomic data</b><br>DOI: 10.1038/s41467-021-25534-2<br>Article page:<br><a href="https://doi.org/10.1038/s41467-021-25534-2">https://doi.org/10.1038/s41467-021-25534-2</a>                                                                                                                                                                                                                                                                                                                         | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> No<br><b>Uses (select all):</b><br><b>Primary use:</b> Mention only<br><b>Biological insight:</b> No<br><b>Evidence sufficiency:</b> Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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|                                                        | <p>Full text 1:<br/> <a href="https://www.nature.com/articles/s41467-021-25534-2.pdf">https://www.nature.com/articles/s41467-021-25534-2.pdf</a><br/> Full text 2:<br/> <a href="https://www.biorxiv.org/content/biorxiv/early/2021/06/10/2021.01.13.426593.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2021/06/10/2021.01.13.426593.full.pdf</a><br/> Controller-registered target occurrence locator(s):<br/> occ-0001<br/> (document_792fe740c44cb4cb3ffb.txt:p-0002);<br/> occ-0002<br/> (document_792fe740c44cb4cb3ffb.txt:p-0014);<br/> occ-0003<br/> (document_1726b16a80b9449cb429.txt:p-0003);<br/> occ-0004<br/> (document_1726b16a80b9449cb429.txt:p-0022)</p> | <p><b>Supporting quote + locator:</b> Lotfollahi et al.<sup>26</sup> developed a VAE model called scGen to infer the expression difference due to perturbation conditions by latent space interpolation.</p> <p><b>Notes:</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>SCGEN-HV-001</b><br/> FORMAL<br/> VALIDATION</p> | <p><b>Opportunities for Artificial Intelligence in Advancing Precision Medicine</b><br/> DOI: 10.1007/s40142-019-00177-4<br/> Article page:<br/> <a href="https://doi.org/10.1007/s40142-019-00177-4">https://doi.org/10.1007/s40142-019-00177-4</a><br/> Full text 1:<br/> <a href="https://link.springer.com/content/pdf/10.1007/s40142-019-00177-4.pdf">https://link.springer.com/content/pdf/10.1007/s40142-019-00177-4.pdf</a><br/> Controller-registered target occurrence locator(s):<br/> occ-0001<br/> (document_d89a75e950e99bb0d716.txt:p-0006)</p>                                                                                                                        | <p><b>Review status:</b><br/> <b>Citation validity:</b> Yes<br/> <b>Method executed:</b> No<br/> <b>Uses (select all):</b><br/> <b>Primary use:</b> Mention only<br/> <b>Biological insight:</b> No<br/> <b>Evidence sufficiency:</b> Yes<br/> <b>Supporting quote + locator:</b><br/> hose primary goal is to establish, discover, and catalogue different cell populations ab initio, creates unsupervised maps, serving as a resource for subsequent disease-directed studies. In addition, it is possible to predict cycle, disease progression, and perturbation responses using deep network approaches [<a href="#">41</a>·, <a href="#">42</a>·, <a href="#">43</a>,<a href="#">41</a>,<a href="#">45</a>].</p> <p><b>Notes:</b></p> |
| <p><b>SCGEN-HV-011</b><br/> FORMAL<br/> VALIDATION</p> | <p><b>A Cross-Modal Autoencoder Framework Learns Holistic Representations of Cardiovascular State</b><br/> DOI: 10.1101/2022.05.26.493497<br/> Article page:<br/> <a href="https://doi.org/10.1101/2022.05.26.493497">https://doi.org/10.1101/2022.05.26.493497</a><br/> Full text 1:<br/> <a href="https://www.biorxiv.org/content/biorxiv/early/2022/05/28/2022.05.26.493497.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2022/05/28/2022.05.26.493497.full.pdf</a><br/> Controller-registered target occurrence locator(s):<br/> occ-0001<br/> (document_15f5b7d997bf75c8406b.txt:p-0011)</p>                                                                           | <p><b>Review status:</b><br/> <b>Citation validity:</b> Yes<br/> <b>Method executed:</b> No<br/> <b>Uses (select all):</b><br/> <b>Primary use:</b> Mention only<br/> <b>Biological insight:</b> No<br/> <b>Evidence sufficiency:</b> Yes<br/> <b>Supporting quote + locator:</b><br/> These models have been successfully applied in a variety of applications including computer vision [<a href="#">5–7</a>], chemistry [<a href="#">8</a>], and biology [<a href="#">9–14</a>].</p> <p><b>Notes:</b></p>                                                                                                                                                                                                                                 |
| <p><b>SCGEN-HV-046</b><br/> FORMAL<br/> VALIDATION</p> | <p><b>Biological network-inspired interpretable variational autoencoder</b><br/> DOI: 10.1101/2020.12.17.423310<br/> Article page:<br/> <a href="https://doi.org/10.1101/2020.12.17.423310">https://doi.org/10.1101/2020.12.17.423310</a><br/> Full text 1:<br/> <a href="https://www.biorxiv.org/content/biorxiv/early/2020/12/19/2020.12.17.423310.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2020/12/19/2020.12.17.423310.full.pdf</a></p>                                                                                                                                                                                                                            | <p><b>Review status:</b><br/> <b>Citation validity:</b> Yes<br/> <b>Method executed:</b> No<br/> <b>Uses (select all):</b><br/> <b>Primary use:</b> Mention only<br/> <b>Biological insight:</b> No<br/> <b>Evidence sufficiency:</b> Yes<br/> <b>Supporting quote + locator:</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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|                                      | <p>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_62ca3a461f8c2418ce78.txt:p-0008);<br/>occ-0002<br/>(document_62ca3a461f8c2418ce78.txt:p-0008);<br/>occ-0003<br/>(document_62ca3a461f8c2418ce78.txt:p-0025);<br/>occ-0004<br/>(document_62ca3a461f8c2418ce78.txt:p-0033);<br/>occ-0005<br/>(document_62ca3a461f8c2418ce78.txt:p-0051);<br/>occ-0006<br/>(document_62ca3a461f8c2418ce78.txt:p-0080)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>More recently, deep generative models such as Variational Autoencoders<sup>7</sup> (VAEs) have proven to be extremely useful for the probabilistic modeling of single-cell transcriptomes, such as scVI or scGen 8–10 .</p> <p><b>Notes:</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| SCGEN-HV-003<br>FORMAL<br>VALIDATION | <p><b>Modelling stem cell differentiation related processes—A practical overview for biologists</b><br/>DOI: 10.1002/1873-3468.70395<br/>Article page:<br/><a href="https://doi.org/10.1002/1873-3468.70395">https://doi.org/10.1002/1873-3468.70395</a><br/>Full text 1:<br/><a href="https://ris.utwente.nl/ws/files/546291702/FEBS_Letters_-_2026_-_Zeegelaar_-_Modelling_stem_cell_differentiation_related_processes_A_practical_overview_for_biologists.pdf">https://ris.utwente.nl/ws/files/546291702/FEBS_Letters_-_2026_-_Zeegelaar_-_Modelling_stem_cell_differentiation_related_processes_A_practical_overview_for_biologists.pdf</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_b87f691ad0b918d74b4f.txt:p-0013);<br/>occ-0002<br/>(document_b87f691ad0b918d74b4f.txt:p-0016);<br/>occ-0003<br/>(document_b87f691ad0b918d74b4f.txt:p-0020)</p> | <p><b>Review status:</b><br/><b>Citation validity:</b> Yes<br/><b>Method executed:</b> No<br/><b>Uses (select all):</b><br/><b>Primary use:</b> Mention only<br/><b>Biological insight:</b> No<br/><b>Evidence sufficiency:</b> Yes<br/><b>Supporting quote + locator:</b><br/>Variational autoencoder-based models, such as scGen, have been used to predict transcriptional responses to gene perturbation by knockout or drug treatment, enabling <i>in silico</i> prioritisation of experimental intervention <a href="#">[[60]]</a>.</p> <p><b>Notes:</b></p>                                                                                                                                                                                                                                       |
| SCGEN-HV-014<br>FORMAL<br>VALIDATION | <p><b>Fluctuation structure predicts genome-wide perturbation outcomes</b><br/>DOI: 10.1101/2025.06.27.661814<br/>Article page:<br/><a href="https://doi.org/10.1101/2025.06.27.661814">https://doi.org/10.1101/2025.06.27.661814</a><br/>Full text 1:<br/><a href="https://www.biorxiv.org/content/biorxiv/early/2025/07/01/2025.06.27.661814.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2025/07/01/2025.06.27.661814.full.pdf</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_8e25f50063871c9d0b5c.txt:p-0007);<br/>occ-0002<br/>(document_8e25f50063871c9d0b5c.txt:p-0082)</p>                                                                                                                                                                                                                                                             | <p><b>Review status:</b><br/><b>Citation validity:</b> Yes<br/><b>Method executed:</b> No<br/><b>Uses (select all):</b><br/><b>Primary use:</b> Mention only<br/><b>Biological insight:</b> No<br/><b>Evidence sufficiency:</b> Yes<br/><b>Supporting quote + locator:</b><br/>Deep learning frameworks <sup>20–22</sup> model perturbation responses in latent spaces and can predict the outcomes of perturbations.</p> <p><b>Notes:</b></p>                                                                                                                                                                                                                                                                                                                                                           |
| SCGEN-HV-041<br>FORMAL<br>VALIDATION | <p><b>A Roadmap for a Consensus Human Skin Cell Atlas and Single-Cell Data Standardization</b><br/>DOI: 10.1016/j.jid.2023.03.1679<br/>Article page:<br/><a href="https://doi.org/10.1016/j.jid.2023.03.1679">https://doi.org/10.1016/j.jid.2023.03.1679</a><br/>Full text 1:<br/><a href="https://uu.diva-portal.org/smash/get/diva2:1832264/FULLTEXT01">https://uu.diva-portal.org/smash/get/diva2:1832264/FULLTEXT01</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_340f03fd4110e8bb63f7.txt:p-0005);<br/>occ-0002<br/>(document_340f03fd4110e8bb63f7.txt:p-0010)</p>                                                                                                                                                                                                                                                                                  | <p><b>Review status:</b><br/><b>Citation validity:</b> Yes<br/><b>Method executed:</b> No<br/><b>Uses (select all):</b><br/><b>Primary use:</b> Mention only<br/><b>Biological insight:</b> No<br/><b>Evidence sufficiency:</b> Yes<br/><b>Supporting quote + locator:</b><br/>Some methods, for example, fastMNN (<a href="#">Haghverdi et al., 2018</a>), scANVI (<a href="#">Xu et al., 2021</a>), and scGen (<a href="#">Lotfollahi et al., 2019</a>), are more effective at ensuring biological conservation, at the risk of retaining some batch effects, whereas others, for example, BBKNN (<a href="#">Polański et al., 2020</a>), Scanorama (<a href="#">Hie et al., 2019</a>), and trVAE (<a href="#">Lotfollahi et al., 2020</a>), are more effective at batch correction at the risk of</p> |

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|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>obscuring biological variation (<a href="#">Luecken et al., 2022</a>).</p> <p>Notes:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| SCGEN-HV-009<br>FORMAL<br>VALIDATION | <p><b>PerTurboAgent: A Self-Planning Agent for Boosting Sequential Perturb-seq Experiments</b><br/>DOI: 10.1101/2025.05.25.656020<br/>Article page:<br/><a href="https://doi.org/10.1101/2025.05.25.656020">https://doi.org/10.1101/2025.05.25.656020</a><br/>Full text 1:<br/><a href="https://www.biorxiv.org/content/biorxiv/early/2025/05/27/2025.05.25.656020.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2025/05/27/2025.05.25.656020.full.pdf</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_902e1d6b1e395793559e.txt:p-0013)</p>                            | <p><b>Review status:</b><br/><b>Citation validity:</b> Yes<br/><b>Method executed:</b> No<br/><b>Uses (select all):</b><br/><b>Primary use:</b> Mention only<br/><b>Biological insight:</b> No<br/><b>Evidence sufficiency:</b> Yes<br/><b>Supporting quote + locator:</b></p> <p>Indeed, knowledge about correlated gene expression patterns and shared gene functions has been leveraged for machine learning models, including generative and foundation models which aim to predict the impact of unseen perturbations (<a href="#">Roohani et al., 2024a</a>; <a href="#">Bunne et al., 2023</a>; <a href="#">Cui et al., 2024</a>; <a href="#">Lotfollahi et al., 2019</a>; 2023;</p> <p>Notes:</p> |
| SCGEN-HV-013<br>FORMAL<br>VALIDATION | <p><b>Biologically Interpretable VAE with Supervision for Transcriptomics Data Under Ordinal Perturbations</b><br/>DOI: 10.1101/2024.03.28.587231<br/>Article page:<br/><a href="https://doi.org/10.1101/2024.03.28.587231">https://doi.org/10.1101/2024.03.28.587231</a><br/>Full text 1:<br/><a href="https://www.biorxiv.org/content/biorxiv/early/2024/03/29/2024.03.28.587231.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2024/03/29/2024.03.28.587231.full.pdf</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_a83cd4b670847c80df1a.txt:p-0008)</p>            | <p><b>Review status:</b><br/><b>Citation validity:</b> Yes<br/><b>Method executed:</b> No<br/><b>Uses (select all):</b><br/><b>Primary use:</b> Mention only<br/><b>Biological insight:</b> No<br/><b>Evidence sufficiency:</b> Yes<br/><b>Supporting quote + locator:</b></p> <p>Exploring and understanding how cellular responses evolve across a spectrum of stimulus doses sheds light on the underlying cellular mechanisms and paves the way for precision medicine approaches tailored to individualized dosedependent responses specifically in the context of disease and drug treatment (<a href="#">Bock et al., 2022</a>; <a href="#">Lotfollahi et al., 2019</a>).</p> <p>Notes:</p>        |
| SCGEN-HV-006<br>FORMAL<br>VALIDATION | <p><b>BuDDI: Bulk Deconvolution with Domain Invariance to predict cell-type-specific perturbations from bulk</b><br/>DOI: 10.1371/journal.pcbi.1012742<br/>Article page:<br/><a href="https://doi.org/10.1371/journal.pcbi.1012742">https://doi.org/10.1371/journal.pcbi.1012742</a><br/>Full text 1:<br/><a href="https://www.biorxiv.org/content/biorxiv/early/2023/10/05/2023.07.20.549951.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2023/10/05/2023.07.20.549951.full.pdf</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_cfff75af3fe5e992064e.txt:p-0015)</p> | <p><b>Review status:</b><br/><b>Citation validity:</b> Yes<br/><b>Method executed:</b> No<br/><b>Uses (select all):</b><br/><b>Primary use:</b> Mention only<br/><b>Biological insight:</b> NO<br/><b>Evidence sufficiency:</b> Yes<br/><b>Supporting quote + locator:</b></p> <p>e cell-type-specific expression profiles across the conditions observed in the bulk data. To accomplish this goal, we build on ideas from three methodological approaches: bulk deconvolution [<a href="#">4–14</a>], variational autoencoder (VAE) [<a href="#">15</a>] models for perturbation prediction [<a href="#">16–22,23</a>], and disentanglement methods [<a href="#">18,24–28</a>].</p> <p>Notes:</p>       |

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| <b>SCGEN-HV-030</b><br>FORMAL<br>VALIDATION | <b>Reconstructing unobserved cellular states from paired single-cell lineage tracing and transcriptomics data</b><br>DOI: 10.1101/2021.05.28.446021<br>Article page:<br><a href="https://doi.org/10.1101/2021.05.28.446021">https://doi.org/10.1101/2021.05.28.446021</a><br>Full text 1:<br><a href="https://www.biorxiv.org/content/biorxiv/early/2021/05/30/2021.05.28.446021.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2021/05/30/2021.05.28.446021.full.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_6d3eff25cfdd9a79809d.txt:p-0008)                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> No<br><b>Uses (select all):</b><br><b>Primary use:</b> Mention only<br><b>Biological insight:</b> NO<br><b>Evidence sufficiency:</b> Yes<br><b>Supporting quote + locator:</b><br>Such heuristics have been used in several scRNA-seq data analysis scenarios [16]. We ran VAE and TreeVAE on a NVIDIA TITAN Xp, and fitting the data took less than a few minutes for every dataset.<br><br><b>Notes:</b>                                                         |
| <b>SCGEN-HV-025</b><br>FORMAL<br>VALIDATION | <b>VI-VS: calibrated identification of feature dependencies in single-cell multiomics</b><br>DOI: 10.1186/s13059-024-03419-z<br>Article page:<br><a href="https://doi.org/10.1186/s13059-024-03419-z">https://doi.org/10.1186/s13059-024-03419-z</a><br>Full text 1:<br><a href="https://genomebiology.biomedcentral.com/counter/pdf/10.1186/s13059-024-03419-z">https://genomebiology.biomedcentral.com/counter/pdf/10.1186/s13059-024-03419-z</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_cfe0d58ca6c7605aef38.txt:p-0035)                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> No<br><b>Uses (select all):</b><br><b>Primary use:</b> Mention only<br><b>Biological insight:</b> No<br><b>Evidence sufficiency:</b> Yes<br><b>Supporting quote + locator:</b><br>In this model, the prior is usually the standard normal, and the posterior distribution is approximated using a variational approach, with the approximation parameterized by neural networks [32, 74, 75].<br><br><b>Notes:</b>                                                 |
| <b>SCGEN-HV-010</b><br>FORMAL<br>VALIDATION | <b>Learning Perturbation-specific Cell Representations for Prediction of Transcriptional Response across Cellular Contexts</b><br>DOI: 10.1101/2023.03.20.533433<br>Article page:<br><a href="https://doi.org/10.1101/2023.03.20.533433">https://doi.org/10.1101/2023.03.20.533433</a><br>Full text 1:<br><a href="https://www.biorxiv.org/content/biorxiv/early/2023/03/23/2023.03.20.533433.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2023/03/23/2023.03.20.533433.full.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_01301fc720e62a99efc8.txt:p-0002);<br>occ-0002<br>(document_01301fc720e62a99efc8.txt:p-0003);<br>occ-0003<br>(document_01301fc720e62a99efc8.txt:p-0008);<br>occ-0004<br>(document_01301fc720e62a99efc8.txt:p-0008);<br>occ-0005<br>(document_01301fc720e62a99efc8.txt:p-0009);<br>occ-0006<br>(document_01301fc720e62a99efc8.txt:p-0009);<br>occ-0007<br>(document_01301fc720e62a99efc8.txt:p-0009);<br>occ-0008<br>(document_01301fc720e62a99efc8.txt:p-0009); | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> Yes<br><b>Uses (select all):</b><br><b>Primary use:</b> Benchmark<br><b>Biological insight:</b> No<br><b>Evidence sufficiency:</b> Yes<br><b>Supporting quote + locator:</b><br>Performance metrics for PerturbX, scGen and the baseline model. $R^2$ results are reported on all cell lines, and separately for <i>TP53</i> -WT and <i>TP53</i> -mutated cell lines. All metrics are measured over the top 100 DEGs unless stated otherwise.<br><br><b>Notes:</b> |

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|                                                 | occ-0009<br>(document_01301fc720e62a99efc8.txt:p-0011);<br>occ-0010<br>(document_01301fc720e62a99efc8.txt:p-0011);<br>occ-0011<br>(document_01301fc720e62a99efc8.txt:p-0011);<br>occ-0012<br>(document_01301fc720e62a99efc8.txt:p-0011);<br>occ-0013<br>(document_01301fc720e62a99efc8.txt:p-0013)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>SCGEN-HV-038</b><br>FORMAL<br>VALIDAT<br>ION | <b>The manatee variational autoencoder model for predicting gene expression alterations caused by transcription factor perturbations</b><br>DOI: 10.1038/s41598-024-62620-z<br>Article page:<br><a href="https://doi.org/10.1038/s41598-024-62620-z">https://doi.org/10.1038/s41598-024-62620-z</a><br>Full text 1:<br><a href="https://www.nature.com/articles/s41598-024-62620-z.pdf">https://www.nature.com/articles/s41598-024-62620-z.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_4bee3cf5dc371f4e90dd.txt:p-0001);<br>occ-0002<br>(document_4bee3cf5dc371f4e90dd.txt:p-0006)                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> No<br><b>Uses (select all):</b><br><b>Primary use:</b> Mention only<br><b>Biological insight:</b> No<br><b>Evidence sufficiency:</b> Yes<br><b>Supporting quote + locator:</b><br>Predicting cellular transcriptomic responses to perturbations remains challenging. scGen <sup>1</sup> VAE and Ghahramani et al. <sup>2</sup><br><br><b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>SCGEN-HV-029</b><br>FORMAL<br>VALIDAT<br>ION | <b>Style transfer with variational autoencoders is a promising approach to RNA-Seq data harmonization and analysis</b><br>DOI: 10.1093/bioinformatics/btaa624<br>Article page:<br><a href="https://doi.org/10.1093/bioinformatics/btaa624">https://doi.org/10.1093/bioinformatics/btaa624</a><br>Full text 1:<br><a href="https://www.biorxiv.org/content/biorxiv/early/2020/04/28/791962.full.pdf">https://www.biorxiv.org/content/biorxiv/early/2020/04/28/791962.full.pdf</a><br>Controller-registered target occurrence locator(s):<br>occ-0001<br>(document_e4c1ddba928d429c11fb.txt:p-0006);<br>occ-0002<br>(document_e4c1ddba928d429c11fb.txt:p-0006);<br>occ-0003<br>(document_e4c1ddba928d429c11fb.txt:p-0006);<br>occ-0004<br>(document_e4c1ddba928d429c11fb.txt:p-0006);<br>occ-0005<br>(document_e4c1ddba928d429c11fb.txt:p-0010);<br>occ-0006<br>(document_e4c1ddba928d429c11fb.txt:p-0010);<br>occ-0007<br>(document_e4c1ddba928d429c11fb.txt:p-0018);<br>occ-0008<br>(document_e4c1ddba928d429c11fb.txt:p-0031) | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> Yes<br><b>Uses (select all):</b><br><b>Primary use:</b> Benchmark<br><b>Biological insight:</b> No<br><b>Evidence sufficiency:</b> Yes<br><b>Supporting quote + locator:</b><br>The data were taken from scGen examples ( <a href="https://github.com/theislabs/scgen-reproducibility">https://github.com/theislabs/scgen-reproducibility</a> ). The dataset was provided by the authors ( <a href="#">Lotfollahi et al., 2019a</a> ) as normalized and log-transformed. The data included 18 868 cells belonging to 8 cellular types and 6998 genes in two conditions. The examples can be found at their project repository: <a href="https://nbviewer.jupyter.org/github/M0hamma dL/scGen_notebooks/blob/master/notebooks/scgen_kang.ipynb">https://nbviewer.jupyter.org/github/M0hamma dL/scGen_notebooks/blob/master/notebooks/scgen_kang.ipynb</a> .<br><br><b>Notes:</b> |
| <b>SCGEN-HV-016</b><br>FORMAL<br>VALIDAT<br>ION | <b>Neural network-assisted humanisation of COVID-19 hamster transcriptomic data reveals matching severity states in human disease</b><br>DOI: 10.1016/j.ebiom.2024.105312<br>Article page:<br><a href="https://doi.org/10.1016/j.ebiom.2024.105312">https://doi.org/10.1016/j.ebiom.2024.105312</a><br>Full text 1:<br><a href="https://edoc.mdc-berlin.de/id/eprint/24800/1/24800oa.pdf">https://edoc.mdc-berlin.de/id/eprint/24800/1/24800oa.pdf</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Review status:</b><br><b>Citation validity:</b> Yes<br><b>Method executed:</b> Yes<br><b>Uses (select all):</b><br><b>Primary use:</b> Apply to Biology Data<br><b>Biological insight:</b> Yes<br><b>Evidence sufficiency:</b> Yes<br><b>Supporting quote + locator:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      | <p>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_a720c35675d8cb1f5507.txt:p-0003);<br/>occ-0002<br/>(document_a720c35675d8cb1f5507.txt:p-0003);<br/>occ-0003<br/>(document_a720c35675d8cb1f5507.txt:p-0003);<br/>occ-0004<br/>(document_a720c35675d8cb1f5507.txt:p-0004);<br/>occ-0005<br/>(document_a720c35675d8cb1f5507.txt:p-0016)</p>                                                                                                                                                                                                                                                                                                                                    | <p>Here, we employ an scGen-based framework to match COVID-19 disease severity levels across humans and hamster species, while also developing a metric for measuring interspecies similarity on a cellular level.</p> <p><b>Notes:</b></p>                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>SCGEN-HV-031</b><br/>FORMAL<br/>VALIDATION</p> | <p><b>Learning genetic perturbation effects with variational causal inference</b><br/>DOI: 10.1371/journal.pcbi.1013194<br/>Article page:<br/><a href="https://doi.org/10.1371/journal.pcbi.1013194">https://doi.org/10.1371/journal.pcbi.1013194</a><br/>Full text 1:<br/><a href="https://journals.plos.org/ploscompbiol/article/file?id=10.1371/journal.pcbi.1013194&amp;type=printable">https://journals.plos.org/ploscompbiol/article/file?id=10.1371/journal.pcbi.1013194&amp;type=printable</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_0d4ecf1c37b93e695e22.txt:p-0017)</p>                                                                                | <p><b>Review status:</b><br/><b>Citation validity:</b> Yes<br/><b>Method executed:</b> No<br/><b>Uses (select all):</b><br/><b>Primary use:</b> Mention only<br/><b>Biological insight:</b> No<br/><b>Evidence sufficiency:</b> Yes<br/><b>Supporting quote + locator:</b><br/>Prior work have also used Gaussian-based latent modeling of single-cell data [30,31]. In other words, Eq (1) is realized by</p> <p><b>Notes:</b></p>                                                                                                                                                                                                     |
| <p><b>SCGEN-HV-023</b><br/>FORMAL<br/>VALIDATION</p> | <p><b>Can Pharmaceutical World Models Become Scientific Instruments? A State-of-the-Art Review of Simulation, Reasoning, and Validation</b><br/>DOI: 10.51847/hn9ejvqjoc<br/>Article page: <a href="https://doi.org/10.51847/hn9ejvqjoc">https://doi.org/10.51847/hn9ejvqjoc</a><br/>Full text 1:<br/><a href="https://pharmacophorejournal.com/storage/files/article/8753dc7a-5010-4f82-ab26-e222c36d7a92-ujOPb8uV7cclw8pM/GdSs6Qs3wBWZkW6.pdf">https://pharmacophorejournal.com/storage/files/article/8753dc7a-5010-4f82-ab26-e222c36d7a92-ujOPb8uV7cclw8pM/GdSs6Qs3wBWZkW6.pdf</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_fc575441ba4c0efdc5f3.txt:p-0025)</p> | <p><b>Review status:</b><br/><b>Citation validity:</b> Yes<br/><b>Method executed:</b> No<br/><b>Uses (select all):</b><br/><b>Primary use:</b> Mention only<br/><b>Biological insight:</b> No<br/><b>Evidence sufficiency:</b> Yes<br/><b>Supporting quote + locator:</b><br/>Perturbation-response models demonstrate that latent cellular representations can be conditioned on interventions and used to predict changes across selected cellular contexts [12].</p> <p><b>Notes:</b></p>                                                                                                                                           |
| <p><b>SCGEN-HV-015</b><br/>FORMAL<br/>VALIDATION</p> | <p><b>A practical guide to sequencing in neuropsychiatric research</b><br/>DOI: 10.1038/s44277-025-00041-0<br/>Article page:<br/><a href="https://doi.org/10.1038/s44277-025-00041-0">https://doi.org/10.1038/s44277-025-00041-0</a><br/>Full text 1:<br/><a href="https://www.nature.com/articles/s44277-025-00041-0.pdf">https://www.nature.com/articles/s44277-025-00041-0.pdf</a><br/>Controller-registered target occurrence locator(s):<br/>occ-0001<br/>(document_93c07b9a77d02618c3c9.txt:p-0008)</p>                                                                                                                                                                                                 | <p><b>Review status:</b><br/><b>Citation validity:</b> Yes<br/><b>Method executed:</b> No<br/><b>Uses (select all):</b><br/><b>Primary use:</b> Mention only<br/><b>Biological insight:</b> No<br/><b>Evidence sufficiency:</b> Yes<br/><b>Supporting quote + locator:</b><br/>seudobulking allows for statistically robust conventional DGE analysis to be performed, but ignores any cellular heterogeneity within a cell type/cluster. While various methods have been developed for DGE analysis in single-cell datasets [88,89,90,91], pseudobulking remains the standard and robustly validated approach</p> <p><b>Notes:</b></p> |